The Fundamental Theorem of Calculus

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: FTC Part 2: evaluating definite integrals**

Evaluate $\int_0^3 (x^2 - 2x)\,dx$ using the Fundamental Theorem of Calculus.

$F(x) = \frac{x^3}{3} - x^2$. $F(3) - F(0) = (9 - 9) - (0) = 0$.

2 **[Foundational]** • **Target: FTC Part 1: accumulation functions**

Let $g(x) = \int_0^x t^2\,dt$. Find $g'(x)$ using FTC Part 1.

By FTC Part 1, $g'(x) = x^2$ (the integrand evaluated at x).

3 **[Foundational]** • **Target: FTC Part 1: accumulation functions**

Let $g(x) = \int_0^{x^2} e^{-t}\,dt$. Find $g'(x)$ using FTC Part 1 combined with the Chain Rule.

$g'(x) = e^{-x^2} \cdot \frac{d}{dx}(x^2) = 2xe^{-x^2}$.

4 **[Foundational]** • **Target: Interpreting accumulation functions**

A particle's velocity is $v(t) = 3 - 2t$ (m/s). Let $s(t) = \int_0^t v(\tau)\,d\tau$ be its displacement from $t = 0$. Find $s(4)$ and interpret its meaning.

$s(4) = \int_0^4 (3 - 2\tau)\,d\tau = [3\tau - \tau^2]_0^4 = (12 - 16) - 0 = -4$. This means the particle's net displacement from $t = 0$ to $t = 4$ is -4 m (it ends up 4 m to the left/behind its starting point).

5 **[Foundational]** • **Target: Interpreting accumulation functions**

Given the graph of f consists of straight-line segments with $f(0) = 0$, and $g(x) = \int_0^x f(t)\,dt$, explain what feature of g's graph occurs where f crosses the x-axis, and where f has a local max.

Where f crosses the x-axis (changes sign), $g' = f$ changes sign, so g has a local extremum there. Where f has a local max, $g' = f$ has a local max, meaning g has an inflection point there (since $g'' = f'$ changes sign from $+$ to $-$).

6 **[Foundational]** • **Target: Average value of a function**

The temperature in a city over a 24-hour period is modeled by $T(t) = 70 + 10\sin\left(\frac{\pi t}{12}\right)$ for $0 \le t \le 24$. Find the average temperature over this period.

Average $= \frac{1}{24}\int_0^{24} \left[70 + 10\sin\left(\frac{\pi t}{12}\right)\right] dt$. The sine term integrates to 0 over a full period (24 hours $= 2$ full cycles), so the average $= 70°$.

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7 [Foundational] • Target: Definite integrals with substitution-friendly integrands

Evaluate $\int_0^1 2x(x^2+1)^3\,dx$ using $u = x^2+1$.

$du = 2x\,dx$. When $x=0$, $u=1$; when $x=1$, $u=2$. $\int_1^2 u^3\,du = \left[\frac{u^4}{4}\right]_1^2 = \frac{16}{4} - \frac{1}{4} = \frac{15}{4}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The Mean Value Theorem for integrals

Find the value c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$, i.e. the point where $f(c)$ equals the average value found earlier.

The average value of $f(x) = x^2$ on $[0, 3]$ is 3 (found earlier). Set $f(c) = c^2 = 3$: $c = \sqrt{3} \approx 1.73$, which lies in $(0, 3)$.

9 [Intermediate] • Target: Visualizing FTC Part 1 as an accumulation function

The graph shows $f(t) = t^2$ (the integrand from $g(x) = \int_0^x t^2\,dt$). Explain, using the graph, why g is always increasing, and why $g'(x) = f(x) = x^2$ matches the height of the curve at each x.

Since $f(t) = t^2 \geq 0$ for all t, every strip added as x increases contributes non-negative area, so $g(x) = \int_0^x t^2\,dt$ is always increasing (or constant only at $x = 0$). By FTC Part 1, $g'(x)$ equals the height of the integrand curve at x, i.e. $g'(x) = f(x) = x^2$, matching the graph's height directly.

10 [Intermediate] • Target: A capstone FTC application

Water flows into a tank at rate $r(t) = 8 - t$ liters/minute for $0 \leq t \leq 8$ (the rate decreases as the tank fills). Using FTC, find the total volume of water added during the first 6 minutes, and explain why the rate being zero at $t = 8$ is consistent with the tank reaching capacity.

Volume $= \int_0^6 (8 - t)\,dt = \left[8t - \frac{t^2}{2}\right]_0^6 = 48 - 18 = 30$ liters. Since $r(t) = 8 - t = 0$ at $t = 8$, the inflow rate drops to zero exactly when the tank presumably reaches capacity, consistent with the water level leveling off.

11 [Intermediate] • Target: FTC Part 1 with more complex bounds

Let $g(x) = \int_3^{2x} \ln(t+1)\,dt$. Find $g'(x)$ using FTC Part 1 and the Chain Rule.

$g'(x) = \ln(2x+1) \cdot \frac{d}{dx}(2x) = 2\ln(2x+1)$.

12 [Intermediate] • Target: A second average-value application

A company's hourly production rate over an 8-hour shift is $p(t) = 50 + 10t - t^2$ units/hour, for $0 \leq t \leq 8$. Find the average production rate over the shift.

Average $= \frac{1}{8}\int_0^8 (50 + 10t - t^2)\,dt = \frac{1}{8}\left[50t + 5t^2 - \frac{t^3}{3}\right]_0^8 = \frac{1}{8}\left(400 + 320 - \frac{512}{3}\right) = \frac{1}{8}\left(\frac{2160 - 512}{3}\right) = \frac{1648}{24} \approx 68.67$ units/hour.

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13 **[Intermediate]** • **Target: Interpreting accumulation functions**

Level 2 · Medium Use Green's Theorem to evaluate $\oint_C (3y\,dx + 5x\,dy)$ where C is the counterclockwise circle $x^2 + y^2 = 4$.

Identify components: $P = 3y$ and $Q = 5x$. Compute partial derivatives: $\frac{\partial Q}{\partial x} = 5$ and $\frac{\partial P}{\partial y} = 3$. Apply Green's Theorem: $\oint_C (3y\,dx + 5x\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA = \iint_R (5-3)\,dA = 2\iint_R dA.$ The region R is a disk of radius $r = 2$, so its area is $\pi(2^2) = 4\pi$. Therefore, $2\iint_R dA = 2(4\pi) = 8\pi$.

14 **[Intermediate]** • **Target: Average value of a function**

[The Fundamental Theorem · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Definite integrals with substitution-friendly integrands** • **Cross-Lesson: 7.01 Differential Equations and Slope Fields**

Solve the equation $z^3 + 8 = 0$, expressing all solutions in Cartesian form $a + bi$.

Write $z^3 = -8 = 8e^{i\pi}$. By De Moivre's Theorem, the three cube roots are:
$z_k = 2\exp\left(i\frac{\pi + 2\pi k}{3}\right)$ for $k = 0, 1, 2$. For $k = 0$: $z_0 = 2e^{i\pi/3} = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 1 + i\sqrt{3}$. For $k = 1$: $z_1 = 2e^{i\pi} = 2(-1) = -2$. For $k = 2$: $z_2 = 2e^{i5\pi/3} = 2e^{-i\pi/3} = 2\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = 1 - i\sqrt{3}$. Notice that $1 \pm i\sqrt{3}$ are complex conjugates, as predicted by the Conjugate Root Theor.

16 **[Challenging]** • **Target: The Mean Value Theorem for integrals** • **Cross-Lesson: 8.01 Area Between Curves**

Phone calls arrive at an emergency center according to a Poisson process with an average rate of 3 calls per minute. Find: (a) the probability that at least 2 calls arrive in a 1-minute period; (b) the probability that exactly 5 calls arrive in a 2-minute period.

(a) For 1 minute, $X \sim \text{Po}(3)$: $P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - e^{-3}\frac{3^0}{0!} - e^{-3}\frac{3^1}{1!} = 1 - e^{-3}(1 + 3) = 1 - 4e^{-3} \approx 1 - 0.19915 = 0.8009 \approx 0.801.$ (b) For 2 minutes, the rate scales linearly: $\lambda = 3 \times 2 = 6$, so $Y \sim \text{Po}(6)$: $P(Y = 5) = \frac{e^{-6} \cdot 6^5}{5!} = \frac{e^{-6} \cdot 7776}{120} = 64.8e^{-6} \approx 0.1606 \approx 0.161.$

17 **[Challenging]** • **Target: Visualizing FTC Part 1 as an accumulation function** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

Level 3 · Advanced Let $f(x)$ be a function with $f(2) = 3$, $f'(2) = -1$, $f''(2) = 4$, and $f'''(2) = -12$. Write the third-degree Taylor polynomial $P_3(x)$ for f centered at $x = 2$, and use it to approximate $f(2.1)$.

Construct the Taylor polynomial using $a = 2$: $P_3(x) = f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3$
$= 3 - 1(x-2) + \frac{4}{2}(x-2)^2 + \frac{-12}{6}(x-2)^3 = 3 - (x-2) + 2(x-2)^2 - 2(x-2)^3$ Substitute $x = 2.1$, so $x - 2 = 0.1$:
$P_3(2.1) = 3 - 0.1 + 2(0.1)^2 - 2(0.1)^3 = 3 - 0.1 + 2(0.01) - 2(0.001)$ $= 3 - 0.1 + 0.02 - 0.002 = 2.918$

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18 [Challenging] • Target: A capstone FTC application • Cross-Lesson: 1.01 Limits Graphically and Numerically

Level 1 · Foundations Find $f_x(2, 1)$ and $f_y(2, 1)$ for the function $f(x, y) = \ln(x^2 + 3y^2)$.

Compute partial derivatives via Chain Rule: $f_x = \frac{2x}{x^2+3y^2}$, $f_y = \frac{6y}{x^2+3y^2}$ Substitute $x = 2$ and $y = 1$: $x^2 + 3y^2 = 2^2 + 3(1^2) = 4 + 3 = 7$ $f_x(2, 1) = \frac{2(2)}{7} = \frac{4}{7}$, $f_y(2, 1) = \frac{6(1)}{7} = \frac{6}{7}$

19 [Challenging] • Target: FTC Part 1 with more complex bounds • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

A continuous random variable X has probability density function $f(x) = 2e^{-2x}$ for $x \geq 0$ and 0 elsewhere. Find the cumulative distribution function $F(x)$ and determine the exact median of X.

1. Find CDF for $x \geq 0$: $F(x) = \int_0^x 2e^{-2t}\,dt = [-e^{-2t}]_0^x = 1 - e^{-2x}$. 2. Set $F(m) = 0.5$ to find the median m: $1 - e^{-2m} = 0.5 \Rightarrow e^{-2m} = 0.5 \Rightarrow -2m = \ln(0.5) = -\ln 2 \Rightarrow m = \frac{\ln 2}{2}$.

20 [Challenging] • Target: A second average-value application • Cross-Lesson: 2.01 The Definition of the Derivative

Level 1 · Foundations Determine whether the vectors $\mathbf{u} = \langle 2, -3, 4 \rangle$ and $\mathbf{v} = \langle 6, 8, 3 \rangle$ are orthogonal, parallel, or neither.

Compute the dot product: $\mathbf{u} \cdot \mathbf{v} = (2)(6) + (-3)(8) + (4)(3) = 12 - 24 + 12 = 0$ Since the dot product is exactly 0 and neither vector is the zero vector, **u** and **v** are strictly orthogonal (perpendicular).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and negative for $t < -2$ and positive for $t > -2$. What feature does g have at $x = -2$?

a. An inflection point, since g is always increasing when its integrand is positive.

b. Neither -- accumulation functions never have local extrema.

c. A local minimum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$. ✓

d. A local maximum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.

By FTC Part 1, $g'(x) = f(x)$. A local minimum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.

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22 [Foundational] • Target: FTC Part 1: accumulation functions

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and positive for $t < 1$ and negative for $t > 1$. What feature does g have at $x = 1$?

a. Neither -- accumulation functions never have local extrema.

b. A local minimum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.

c. A local maximum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$. ✓

d. An inflection point, since g is always increasing when its integrand is positive.

By FTC Part 1, $g'(x) = f(x)$. A local maximum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.

23 [Foundational] • Target: Interpreting accumulation functions

For $F(x) = \int_a^x f(t)dt$ with f continuous, the proof of FTC Part 1 uses the Mean Value Theorem for Integrals to write $F(x + h) - F(x)$ as:

a. hf'(x)

b. $f(c) \cdot h$ for some $c \in [x, x + h]$ ✓

c. $f(x) \cdot (x + h)$

d. f(x+h)-f(x)

MVT for integrals gives $\int_x^{x+h} f(t)dt = f(c)h$ for some c between x and $x + h$; letting $h \to 0$, $c \to x$.

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24 [Foundational] • Target: Average value of a function

If f has a jump discontinuity at $c \in (a, b)$, the accumulation function $F(x) = \int_a^x f(t)dt$ is still:

a. continuous at c (though not differentiable there) ✓

b. discontinuous at c

c. differentiable at c

d. undefined at c

Integrating a bounded (even discontinuous) function always yields a continuous accumulation function; differentiability can fail at the jump.

25 [Foundational] • Target: Definite integrals with substitution-friendly integrands

FTC Part 2: $\int_a^b f(x)dx = F(b) - F(a)$ where $F' = f$. Evaluate $\int_0^{\pi} \cos x \, dx$:

a. 0 ✓

b. 2

c. -2

d. 1

Applying Fundamental Theorem of Calculus principles, $[\sin x]_0^{\pi} = 0 - 0 = 0$. Therefore, the verified correct option is 0.

26 [Foundational] • Target: The Mean Value Theorem for integrals

If f is continuous on $[a, b]$ and F is any antiderivative, FTC Part 2 requires no assumption that F is:

a. equal to f

b. unique – – any antiderivative works, since they differ by a constant ✓

c. bounded on $[a, b]$

d. differentiable everywhere

Any two antiderivatives of f differ by a constant C, which cancels in $F(b) - F(a)$.

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27 [Foundational] • Target: Visualizing FTC Part 1 as an accumulation function

FTC Part 1 shows every continuous function has an:

a. **antiderivative ✓**

b. derivative everywhere

c. limit

d. inflection point

Constructs antiderivative $G(x) = \int_a^x f\,dt$.

28 [Foundational] • Target: A capstone FTC application

Mean value theorem for integrals: $\int_a^b f\,dx = f(c)(b-a)$ for some $c \in (a, b)$. Requires f to be:

a. positive

b. integrable and bounded

c. **continuous ✓**

d. analytic

MVT for integrals: f continuous on $[a, b]$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: FTC Part 1 with more complex bounds

Mean Value Theorem for Integrals: if f is continuous on $[a, b]$, there exists $c \in (a, b)$ with $\int_a^b f(x)\,dx =$

a. $f(c)(b-a)$ ✓

b. $f'(c)(b-a)$

c. $f(b) - f(a)$

d. $f(c)$

This is the integral form of the MVT, closely tied to FTC Part 1: the average value $f(c)$ times the interval length equals the integral.

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30 **[Intermediate]** • **Target: A second average-value application**

For f continuous on $[a, b]$, the function $F(x) = \int_a^x f(t)dt$ is guaranteed to be:

a. differentiable on (a, b), with $F' = f$ ✓

b. only continuous, not differentiable

c. a polynomial

d. constant

FTC Part 1 guarantees F is differentiable wherever f is continuous, with $F'(x) = f(x)$.

31 **[Intermediate]** • **Target: FTC Part 2: evaluating definite integrals**

Proof of FTC: $F(x) = \int_a^x f$. Key step: $F(x+h) - F(x) = \int_x^{x+h} f(t)dt \approx f(x) \cdot h$ by:

a. **MVT for integrals** ✓

b. L'Hôpital

c. product rule

d. chain rule

MVT for integrals gives $\approx f(\xi) \cdot h \to f(x) \cdot h$.

32 **[Intermediate]** • **Target: FTC Part 1: accumulation functions**

FTC Part 1 is proved by constructing:

a. any antiderivative

b. $G(x) = \int_a^x f\,dt$ **and showing** $G' = f$ ✓

c. Taylor series

d. Riemann sums

Key step: $\lim_{h \to 0} \frac{G(x+h) - G(x)}{h} = f(x)$.

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33 [Intermediate] • Target: Interpreting accumulation functions

FTC in higher dimensions: $\int_a^b ablaf \cdot d\mathbf{r} = f(b) - f(a)$. This is valid for:

a. any vector field

b. **gradient fields (conservative) only** ✓

c. curl-free fields on simply-connected domains

d. divergence-free fields

FTC for line integrals: valid for gradient fields.

34 [Intermediate] • Target: Average value of a function

Gradient theorem (FTC for line integrals): $\int_C ablaf \cdot d\mathbf{r} =$

a. $\iint f dA$

b. $f(\mathbf{r}(b)) - f(\mathbf{r}(a))$ ✓

c. $\oint f ds$

d. $abla^2 f$

Path-independence for gradient fields.

35 [Intermediate] • Target: Definite integrals with substitution-friendly integrands

Let $F(x) = \int_2^x (t^2 - 1)dt$. According to the Fundamental Theorem of Calculus, what is the derivative of $F(x)$ with respect to x?

a. $x^2 - 1$ ✓

b. x^2

c. $x^2 - 1 - (2^2 - 1)$

d. $2x$

The Fundamental Theorem of Calculus, Part 1, states that if $F(x) = \int_a^x f(t)dt$, then $F'(x) = f(x)$. In this problem, $f(t) = t^2 - 1$ and the lower limit of integration is $a = 2$. Therefore, $F'(x) = x^2 - 1$.

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36 [Intermediate] • Target: The Mean Value Theorem for integrals

IBP formula follows from FTC because $\int_a^b fg'\,dx = [fg]_a^b - \int_a^b f'g\,dx$ derives from:

a. Differentiating fg and applying FTC Part 2 ✓

b. L'Hôpital

c. Green's theorem

d. Cauchy-Schwarz

Applying Fundamental Theorem principles, Product rule + FTC Part 2. Therefore, the verified correct option is Differentiating fg and applying FTC Part 2.

37 [Intermediate] • Target: Visualizing FTC Part 1 as an accumulation function

FTC allows computing $\int_0^1 x^{100}\,dx$ via:

a. L'Hôpital

b. power rule antiderivative ✓

c. IBP

d. squeeze theorem

Applying Fundamental Theorem principles, Antiderivative $x^{101}/101$. Therefore, the verified correct option is power rule antiderivative.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A capstone FTC application • Cross-Lesson: 1.01 Limits Graphically and Numerically

Average value of f on $[a, b]$:

a. $\frac{f(a)+f(b)}{2}$

b. $\frac{1}{b-a}\int_a^b f\,dx$ ✓

c. $\int_a^b f\,dx$

d. $f\left(\frac{a+b}{2}\right)$

Applying Fundamental Theorem principles, Average value formula. Therefore, the verified correct option is $\frac{1}{b-a}\int_a^b f\,dx$.

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39 [Challenging] • Target: FTC Part 1 with more complex bounds • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Fubini's theorem allows interchanging iterated integrals when:

a. both integrals converge

b. f is continuous

c. $\iint |f| < \infty$ ✓

d. $f \geq 0$

Applying Fundamental Theorem principles, Absolute integrability. Therefore, the verified correct option is $\iint |f| < \infty$.

40 [Challenging] • Target: A second average-value application • Cross-Lesson: 2.01 The Definition of the Derivative

Net change theorem: $\int_1^4 v(t)dt$ gives the net displacement if $v(t)$ is velocity. For $v = 2t$:

a. 15 ✓

b. 8

c. 30

d. 4

Applying Fundamental Theorem of Calculus principles, $[t^2]_1^4 = 16 - 1 = 15$. Therefore, the verified correct option is 15.

41 [Challenging] • Target: FTC Part 2: evaluating definite integrals • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Stokes' theorem $\iint_S (abla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{r}$ generalises FTC to:

a. $\mathbb{R}^3$ surfaces

b. scalar fields

c. **differential forms on manifolds** ✓

d. $\mathbb{R}^2$ only

Stokes/Green/Divergence are all cases of the general Stokes theorem for differential forms.

42 [Challenging] • Target: FTC Part 1: accumulation functions • Cross-Lesson: 3.01 The Chain Rule

The Net Change Theorem $\int_a^b F'(x)dx = F(b) - F(a)$ requires F' to be:

a. continuous and positive

b. integrable on $[a, b]$ ✓

c. a polynomial

d. bounded and increasing

The theorem (a restatement of FTC Part 2) only needs F' integrable on $[a, b]$.

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43 [Challenging] • Target: Interpreting accumulation functions • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

A continuous function f and a constant a satisfy $\int_a^x f(t)\,dt = e^x - e^5$ for all x, with $a > 0$. Find a.

a. 25

b. -5

c. 10

d. 5 ✓

Setting $x = a$ makes the integral 0, so $e^a - e^5 = 0$, giving $a = 5$.

44 [Challenging] • Target: Average value of a function • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

MVT for integrals: $f(x) = x^2$ on $[0, 3]$. Value of c:

a. $\sqrt{3}$ ✓

b. $3/\sqrt{3}$

c. 3/2

d. $\sqrt{3}/3$

$f(c) = \text{avg} = 3 \Rightarrow c^2 = 3 \Rightarrow c = \sqrt{3}$.

45 [Challenging] • Target: Definite integrals with substitution-friendly integrands • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

MVT for integrals guarantees a value c called a:

a. critical point

b. root

c. **mean point** ✓

d. zero

$f(c) = \frac{1}{b-a}\int_a^b f$: average (mean) value.

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